

Notes: Nofsinger and Sias (JF, 1999)

Herding and Feedback Trading by Institutional and Individual Investors

1. Big Picture

This paper studies whether institutional and individual investors herd and feedback-trade, and whether such behavior is connected to stock returns. The central empirical object is the relation between changes in institutional ownership and stock returns. Because individual ownership is the complement of institutional ownership, a rise in institutional ownership is simultaneously a fall in individual ownership.

The main finding is a strong positive association between changes in institutional ownership and contemporaneous returns. Stocks with large increases in institutional ownership earn very high abnormal returns during the same period, while stocks with large decreases in institutional ownership earn very low abnormal returns. The paper interprets this as evidence that either institutions positive-feedback trade more than individuals, or institutional herding has larger price impact than individual herding, or both.

A key conclusion is that this relation does not look like purely destabilizing, irrational institutional herding. The paper finds no short-run post-herding return reversal. Instead, stocks institutions buy continue to outperform stocks they sell. Institutional trading is also linked to return momentum: institutions tend to buy past winners and sell past losers, and stocks that continue to attract institutions display stronger momentum.

2. Research Question

The paper asks four related questions:

1. Are changes in institutional ownership positively or negatively related to stock returns during the same period?
2. Do stocks heavily bought by institutions subsequently reverse, as would be expected if institutional herding pushes prices away from fundamentals?
3. Is institutional herding related to lagged returns, that is, do institutional investors positive-feedback trade?
4. Can the positive relation between institutional ownership changes and returns be decomposed into price impact versus within-period feedback trading?

3. Incremental Contribution

- **Combines two literatures.** Earlier work often treats institutional herding and individual-investor sentiment separately. This paper compares the relative importance of institutional versus individual investor herding by using changes in institutional ownership as a direct aggregate trading measure.
- **Broad NYSE ownership panel.** The paper studies all eligible NYSE firms from 1977 to 1996, rather than focusing only on mutual funds or pension funds.
- **Post-herding tests.** The paper asks whether herding-year returns reverse, continue, or disappear in the years after large ownership changes.
- **Feedback trading and momentum.** The paper connects institutional ownership changes to lag returns and return momentum, rather than treating herding as isolated from standard momentum patterns.
- **Transaction-level evidence.** Using TORQ audit-trail data, the paper examines daily institutional trading and daily returns, providing exploratory evidence on whether contemporaneous price movements are more important than lagged-return feedback over very short horizons.

4. Definitions and Concepts

Herding. Herding is a group of investors trading in the same direction over a period of time. In this paper, the herding interval is usually one year, measured from October through September.

Feedback trading. Feedback trading is a special case of herding in which current trading is correlated with lagged returns. Positive-feedback trading means buying after price increases and selling after price declines.

Institutional versus individual herding. Because the paper observes the fraction of shares owned by institutions, a rise in institutional ownership means institutional buying relative to individual investors. A positive relation between institutional ownership changes and contemporaneous returns indicates that institutions either buy during rising prices more than individuals, or their buying pressure has more price impact.

Abnormal returns. The paper mainly uses monthly abnormal returns computed relative to firms in the same capitalization decile and compounds those monthly abnormal returns over the relevant interval.

5. Data

- **Sample period:** 1977-1996.

- **Universe:** NYSE firms, excluding closed-end funds, REITs, primes and scores, and foreign companies.
- **Returns and size:** monthly CRSP returns and annual market capitalization measured at the beginning of October.
- **Institutional ownership:** annual fraction of shares held by institutions from Standard & Poor’s *Security Owners’ Stock Guides*, interpreted as third-quarter holdings.
- **Main sample size:** 24,869 firm-years with ownership, return, and capitalization data.
- **TORQ sample:** Trades, Orders, Reports, and Quotes data for 144 NYSE stocks from November 1, 1990 to January 31, 1991, with trader type identified for institutional and individual investors. After exclusions, the paper uses 114 firms and focuses on 40 firms with large institutional ownership changes.

6. Empirical Design

Annual ownership-change portfolios. Each October, the paper first sorts firms into 10 portfolios by initial institutional ownership. Within each initial-ownership decile, it then sorts firms into 10 portfolios by the change in institutional ownership over the next year. Finally, it reaggregates across initial-ownership deciles to form 10 ownership-change portfolios. This produces portfolios with similar starting institutional ownership but very different institutional ownership changes.

The institutional ownership change is adjusted for the market-wide annual change:

$$\Delta IO_i = (IO_{i,t+1} - IO_{i,t}) - \overline{(IO_{t+1} - IO_t)}.$$

Post-herding returns. The paper examines returns during the herding year, the first year after herding, and the second year after herding. The key test is whether the high institutional-buying portfolio reverses after its strong contemporaneous return.

Momentum controls. The paper uses independent sorts and Fama-MacBeth regressions to compare the predictive power of institutional ownership changes and lag returns. A representative regression is

$$R_{t=12 \text{ to } 23} = 0.1120 + 0.0427 \cdot \Delta IO \text{ Rank}_{t=0 \text{ to } 11} + 0.0685 \cdot R \text{ Rank}_{t=0 \text{ to } 11}.$$

The institutional-ownership rank coefficient is statistically significant in the paper, suggesting ownership changes help forecast returns even after controlling for momentum.

TORQ daily test. With TORQ data, daily institutional ownership change is measured as

$$\Delta IO_{i,t}^{daily} = \frac{\text{Institutional Buy Volume}_{i,t} - \text{Institutional Sell Volume}_{i,t}}{\text{Shares Outstanding}_i}.$$

The paper regresses daily institutional ownership changes on contemporaneous returns and lagged returns. If institutions are mainly feedback-trading over days, lag returns should

dominate. If institutional trading moves prices or reacts within the day, contemporaneous returns should dominate.

7. Main Findings

Finding 1: institutional ownership changes and contemporaneous returns are strongly positively related. Table I shows that stocks with the largest decrease in institutional ownership lose 13.12 percent abnormal return during the herding year, while stocks with the largest increase gain 18.38 percent. The spread is more than 31 percent per year. This is the paper’s central fact.

Finding 2: no evidence of post-herding reversal. If institutional herding simply pushed prices away from fundamentals, one might expect reversal after the herding year. The paper does not find that. In the first post-herding year, the large-increase portfolio earns 3.05 percent, while the large-decrease portfolio earns -2.38 percent. In the second post-herding year, the return spread is weak. This suggests institutional herding is not obviously destabilizing over a one- to two-year horizon.

Finding 3: institutional ownership changes predict returns beyond standard momentum. Table II and the Fama-MacBeth regression suggest that changes in institutional ownership help predict future returns even when past returns are controlled for. This means institutional trading is not simply a mechanical relabeling of momentum portfolios.

Finding 4: institutional feedback trading is present. Table I’s pre-herding returns show that stocks experiencing future increases in institutional ownership already have positive abnormal returns before the herding year, while stocks institutions later sell have negative lag abnormal returns. This is consistent with positive-feedback trading by institutions.

Finding 5: stock characteristics do not fully explain feedback trading. Table IV examines whether institutions only appear to feedback-trade because they are slowly adjusting toward stocks with desirable characteristics, such as size, liquidity, price, and volatility. The evidence partly supports this for winners, but not for losers. Losers start close to expected institutional ownership and then become under-owned as institutions sell them.

Finding 6: the relation between institutional ownership and momentum is strong. Table V shows that among past winners, those with subsequent institutional selling reverse sharply, while those with subsequent institutional buying continue strongly. Among past losers, those with institutional selling continue to fall, while those with institutional buying rebound. This suggests that return momentum is tightly linked to institutional ownership changes, though causality remains ambiguous.

Finding 7: institutional feedback trading is concentrated in small stocks, but contemporaneous ownership-return comovement exists across sizes. Table VI shows that pre-herding feedback patterns are much stronger in small stocks. However, the herding-year relation between institutional ownership changes and returns is strong across all size

deciles. Even among the largest stocks, institutional-buy stocks outperform institutional-sell stocks substantially during the herding year. This supports both feedback trading and price-pressure interpretations.

Finding 8: TORQ evidence points to contemporaneous effects. Table VII shows that in the three-month TORQ sample, stocks with large institutional ownership increases earn 18.83 percent abnormal returns, while stocks with large decreases earn -9.96 percent. Daily regressions show that daily institutional ownership changes are strongly related to same-day returns but weakly related to lag returns. This is consistent with price impact or very short-horizon feedback trading.

8. Economic Interpretation

The paper’s evidence supports a nuanced interpretation. Institutional investors do positive-feedback trade, especially in smaller stocks. But the strong ownership-return relation cannot be explained solely by slow feedback trading. The evidence from large stocks and TORQ data suggests institutional trades may also move prices, or institutions may react to information within the same day.

The lack of post-herding reversal is important. It implies that institutional buying does not look like simple irrational buying pressure that quickly reverses. Instead, institutional investors appear to buy stocks that continue to do well, at least over the next year. That is consistent with informed trading, with institutions responding to persistent information, or with institutional trading contributing to momentum.

9. Relation to the Literature

- Relative to individual-sentiment models, the paper shows that institutions play an important role in the trading-return relation.
- Relative to mutual-fund performance studies, the paper differs because it studies the returns of assets bought and sold by institutions, not the returns actually earned by institutional investors after costs, fees, and timing.
- Relative to Lakonishok, Shleifer, and Vishny (1992), the paper uses a broader institutional ownership sample and finds stronger evidence of institutional ownership changes being linked to returns.
- Relative to Jegadeesh and Titman (1993), the paper links momentum patterns to institutional ownership changes, showing that momentum and institutional trading are closely connected.

10. Key Limitations

- **Annual ownership data are coarse.** The main institutional ownership data are observed annually, so the paper cannot precisely date institutional trades within the year.
- **Changes in ownership may be concentrated.** A large change in institutional ownership could sometimes reflect a few large institutions rather than broad-based herding.
- **Causality is difficult.** The paper cannot fully distinguish whether institutional trading causes momentum or institutions chase momentum.
- **TORQ evidence is exploratory.** The transaction-level sample covers only three months and a relatively small number of NYSE stocks.
- **Risk adjustment is imperfect.** The paper uses many robustness checks, but any long-horizon abnormal return study faces concerns about risk adjustment and benchmark selection.

11. Table-by-Table Guide

Table I. Establishes the main fact. Institutional buying is strongly positively associated with herding-year returns. It also shows no reversal in the next two years and shows positive lag returns before institutional buying.

Table II. Shows that post-herding returns are not simply momentum returns. Institutional ownership changes remain informative even when stocks are grouped by prior performance.

Table III. Demonstrates robustness of post-herding results to different abnormal return methods: book-to-market adjustment, size adjustment, market adjustment, buy-and-hold returns, and Barber-Lyon matched-firm returns.

Table IV. Connects institutional ownership changes to momentum portfolios. Past winners receive more institutional ownership, past losers lose institutional ownership, and characteristic-adjusted ownership changes only partly explain the effect.

Table V. Shows that among past winners and losers, future returns are strongly related to subsequent changes in institutional ownership. This is one of the strongest pieces of evidence linking institutional trading and momentum.

Table VI. Shows that feedback trading is primarily a small-stock phenomenon, while the same-year ownership-return relation appears in all size groups.

Table VII. Uses TORQ data to show that institutional ownership changes are related to same-day returns more than lagged returns, suggesting contemporaneous price impact or very short-run feedback trading.

12. Extracted Tables from the Paper

	Large Decrease	Decile 2	Decile 3	Decile 4	Decile 5	Decile 6	Decile 7	Decile 8	Decile 9	Large Increase	F-statistic
Panel A: Institutional Ownership Statistics											
Initial % Inst.	0.3762	0.3701	0.3674	0.3651	0.3658	0.3658	0.3669	0.3672	0.3663	0.3642	0.07
Abnstitutional	-0.1595	-0.0714	-0.0418	-0.0247	-0.0122	0.0021	0.0169	0.0365	0.0685	0.1630	624.8***
In/Capital	22.1461	12.3957	12.8506	12.0262	12.0644	12.0419	12.0149	12.2755	12.7228	12.6465	6.30***
In/Book.Mkt.)	-0.3318	-0.4143	-0.4197	-0.4549	-0.4596	-0.4273	-0.4563	-0.4246	-0.4087	-0.4778	0.43
Panel B: Herding Year Abnormal Returns											
t = 0 to 11	-0.1312	-0.0822	-0.0599	-0.0286	-0.0144	-0.0079	0.0112	0.0428	0.0836	0.1838	68.94**
	(-10.86)***	(-7.72)***	(-6.44)***	(-4.44)***	(-3.52)	(-2.03)	(1.70)	(5.56)**	(7.17)**	(9.61)**	
Panel C: Post-Herding Year Abnormal Returns											
t = 12 to 23	-0.0028	-0.0016	-0.0123	-0.0013	0.0013	0.0079	0.0010	0.0141	0.0104	0.0005	3.59***
t = 24 to 25	-1.78**	-2.51***	-1.59	-0.14	-0.13	-0.13	-1.80*	-1.80	-1.80*	-2.48**	0.51
	(-0.96)	(-0.46)	(-0.77)	(0.82)	(1.04)	(0.36)	(0.84)	(-0.45)	(1.23)	(0.86)	
Panel D: Pre-Herding Year Abnormal Returns											
t = -1 to -3	-0.0247	-0.0107	-0.0094	-0.0057	-0.0038	-0.0030	0.0021	-0.0011	0.0101	0.0212	8.96***
t = -1 to -12	-0.025*	-0.033**	-0.031**	-0.033*	-0.033*	-0.033*	-0.033*	-0.033*	-0.033*	-0.033*	3.92***
	(-2.26)**	(-3.32)	(-3.87)	(-3.89)	(-3.73)	(-3.39)	(-3.11)	(-3.13)	(1.42)	(3.61)***	

***, **, and * Statistically significant at the 1, 5, and 10 percent levels, respectively.

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Table I. Characteristics of Institutional Ownership-Change Portfolios.

Abnormal Return	Large Decrease	Decile 2	Decile 3	Decile 4	Decile 5	Decile 6	Decile 7	Decile 8	Decile 9	Large Increase	F-stat.
Panel A: Portfolio Adjusted Returns											
Book/Market decile CAR	-0.0213	-0.0232	-0.0157	-0.0035	0.0097	-0.0009	0.0137	0.0162	0.0101	0.0375	4.58***
	(-1.92*)	(-2.01)**	(-1.36)*	(-0.38)	(1.87)**	(-0.14)	(1.38)	(1.61)	(1.12)	(3.36)***	
Capitalization and B/M decile CAR	-0.0223	-0.0232	-0.0179	-0.0013	0.0086	-0.0009	0.0124	0.0139	0.0138	0.0351	5.40***
	(-1.95**)	(-2.06)**	(-1.42)**	(-0.17)	(1.63)	(-0.14)	(1.56)*	(1.61)	(1.67)*	(3.46)***	
EW market-adjusted CAR	-0.0225	-0.0218	-0.0130	-0.0003	0.0051	-0.0019	0.0104	0.0099	0.0091	0.0258	3.06**
	(-1.38)	(-1.80)*	(-1.50)	(-0.04)	(0.78)	(-0.25)	(1.25)	(1.07)	(0.95)	(1.84)*	
Capitalization decile B&H	-0.0274	-0.0281	-0.0101	-0.0015	0.0032	-0.0021	0.0139	0.0134	0.0128	0.0208	3.63***
	(-2.06)*	(-2.77)**	(-2.39)**	(-0.15)	(0.75)	(-0.39)	(1.81)*	(1.46)	(1.55)	(1.86)*	
Book/Market decile B&H	-0.0199	-0.0204	-0.0168	-0.0087	0.0130	-0.0009	0.0151	0.0175	0.0091	0.0412	4.99***
	(-1.82*)	(-2.04)**	(-2.34)**	(-0.87)	(2.39)**	(-0.12)	(1.36)	(1.79)**	(1.14)	(3.66)**	
Capitalization and B/M decile B&H	-0.0250	-0.0260	-0.0229	-0.0018	0.0081	-0.0055	0.0123	0.0138	0.0123	0.0353	5.13***
	(-2.55)**	(-2.69)**	(-2.54)***	(-0.20)	(1.24)	(-0.75)	(1.77)*	(1.41)	(1.70)	(3.67)**	
EW market-adjusted B&H	-0.0257	-0.0251	-0.0150	-0.0009	0.0055	-0.0030	0.0115	0.0118	0.0082	0.0302	2.55**
	(-1.58)	(-2.30)**	(-1.81)*	(-0.09)	(0.77)	(-0.40)	(1.33)	(1.23)	(0.86)	(1.72)	
Panel B: Matched-Firm Adjusted Returns											
Barber & Lyon abnormal return	-0.0178									0.0400	6.81***
	(-1.26)									(2.96)**	

***, **, and * indicate statistical significance at the 1, 5, and 10 percent levels, respectively.

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Table III. Post-Herding Returns: Alternative Methodologies.

Table V						
Winners and Losers Sorted by Subsequent Changes in Institutional Ownership						
At the beginning of each October (1977-1990), NYSE firms are sorted into 10 portfolios based on their raw performance over the previous six months ($t = -1$ to -6). Panel A reports the time-series average of the annual cross-sectional average subsequent changes in fractional institutional ownership and abnormal returns over the following year ($t = 0$ to 11) for stocks in the top lag ($t = -1$ to -6) performance decile (winners) sorted into subsequent ($t = 0$ to 11) changes in institutional ownership quintiles. Similarly, Panel B reports the data for lag losers sorted into subsequent change in ownership quintiles. Abnormal returns are computed by compensating monthly capitalization decile adjusted returns. F-statistics in parentheses are calculated from time-series standard errors of annual cross-sectional averages. The F-statistic is based on the null hypothesis that the time-series averages of cross-sectional means do not differ across the portfolios.						
	Subsequent Decline in Institutional Ownership	2	3	4	Subsequent Increase in Institutional Ownership	F-statistic
Panel A: Winners Sorted by Subsequent Changes in Institutional Ownership						
$\Delta \text{Inst. (} t = 0 \text{ to } 11)$	-0.1331	-0.0222	0.0003	0.0382	0.1590	337.73***
Subsequent abnormal return ($t = 0$ to 11)	-0.0648	-0.0637	-0.0140	0.0829	0.1919	18.50***
	(-2.01)**	(-2.56)**	(-0.62)	(0.40)***	(5.69)***	
Panel B: Losers Sorted by Subsequent Changes in Institutional Ownership						
$\Delta \text{Inst. (} t = 0 \text{ to } 11)$	-0.1480	-0.0533	-0.0198	0.0123	0.1076	250.36***
Subsequent abnormal return ($t = 0$ to 11)	-0.2355	-0.1395	-0.0445	-0.0066	0.1411	25.30***
	(-6.60)***	(-4.61)***	(-1.26)	(-0.40)	(3.41)**	
***, **, and * Statistically significant at the 1%, 5, and 10 percent levels, respectively.						

***, **, and * Statistically significant at the 1, 5, and 10 percent levels, respectively.

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Panel A: Post-Herding Returns for Stocks Sorted on Herding Year Return and Changes in Institutional Ownership											
Abnstitutional Ownership	Lower	Quintile 2	Quintile 3	Quintile 4	Quintile 5	Winners	F-statistic	Win. Los. F-statistic			
Decrease	-0.0631	0.0061	-0.0141	0.0135	-0.0229	3.28***	0.0402	0.0402	1.39	0.058	
Quintile 2	-0.0402	-0.0216	0.0062	0.0151	0.0206	4.39***	0.0216	0.0216	0.81	0.018	
Quintile 4	-0.0192	-0.0061	0.0080	0.0229	0.0003	0.82	0.014	0.014	0.018	0.018	
Quintile 5	-0.0207	0.0139	0.0122	0.0196	0.0207	1.48	0.018	0.018	1.79*	0.025	
Increase	-0.0051	0.0040	-0.0047	0.0421	0.0074	2.70**	0.014	0.014	1.62	0.018	
F-statistic	1.48	0.97	1.41	1.92	1.92						
In-De. F-statistic	0.0087	0.0083	0.0087	0.0083	0.0083						
	(2.16)**	(-0.13)	(0.72)	(1.55)	(1.31)***						
Panel B: Post-Herding Abnormal Returns by Firm Size for Months 12-33											
Large Increase	Decile 2	Decile 3	Decile 4	Decile 5	Decile 6	Decile 7	Decile 8	Decile 9	Large Increase	F-statistic	
Large firms (log > median)	(-0.90)	(-3.06)***	(-2.25)**	(-0.01)	(-0.40)	(-0.74)	(2.09)**	(2.01)*	(2.40)**	(1.12)	2.97***
Small firms (log < median)	-0.0254	-0.0153	-0.0146	0.0043	0.0115	0.0083	0.0136	0.0136	0.0085	0.0478	2.09**
	(-1.45)	(-1.29)	(-0.89)	(0.28)	(1.65)	(0.83)	(1.10)	(0.86)	(0.79)	(2.71)**	

***, **, and * indicate statistical significance at the 1, 5, and 10 percent levels, respectively.

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Table II. Analyses of Post-Herding Returns.

Table IV											
Momentum Portfolios and Subsequent Changes in Institutional Ownership											
At the beginning of each October (1977-1990), NYSE firms are sorted into 10 portfolios based on their raw performance over the previous six months (t = -1 to -6). Panel A reports the time-series average of the annual cross-sectional mean raw returns for the formation period (t = -1 to -6), abnormal returns over the subsequent 12 months (t = 0 to 11), and the change in the fraction of shares held by institutional investors over the subsequent 12 months. Abnormal returns for each firm are computed by compensating monthly capitalization decile adjusted returns. Abnstitutional is the change in institutional ownership for each firm. Panel B reports the time-series average of the annual cross-sectional mean abnormal levels of and changes in, institutional ownership for firms in each of the 10 lag performance sorted portfolios. The "beginning" abnormal level of institutional ownership is calculated as the residual in a regression of the fraction of shares held by institutional investors immediately following the formation period (i.e., at the beginning of month t = 0) on the firm's characteristics (measured at time t = 0): share price (measured as log t = share price), return standard deviation based on monthly returns over the previous 24 to 60 months depending on availability, return variance (based on monthly returns over the previous 24 to 60 months depending on availability), liquidity (measured as log t = September volume/number of shares outstanding), firm size (measured as log t = equity capitalization), and firm size squared (measured as log t = equity capitalization ²). Similarly, the "ending" abnormal level of institutional ownership is calculated as the residual in a regression of the fractional institutional ownership one year following formation (i.e., at the end of month t = 11) on the same characteristics (measured at t = 0). The abnormal change in institutional ownership is estimated as the difference in these residuals. The F-statistic is based on the null hypothesis that the time-series averages of cross-sectional means do not differ across the portfolios.											
	Lower	Decile 2	Decile 3	Decile 4	Decile 5	Decile 6	Decile 7	Decile 8	Decile 9	Winners	F-statistic
Formation period raw returns (t = -1 to -6)	-0.3939	-1.1482	-0.0089	-0.0128	0.0322	0.078	0.106	0.103	0.268	0.520	53.77***
Subsequent abnormal returns (t = 0 to 11)	-0.0565	-0.0204	-0.0009	-0.0062	-0.0027	0.0041	0.0130	0.0137	0.0191	0.0095	3.26**
ΔInst. (t = 0 to 11)	-1.848**	(-1.44)	(-0.20)	(-1.44)	(0.57)	(0.53)	(1.68)	(1.50)	(1.42)	(1.26)	10.33**
	(-2.09)**	(-0.91)	(-0.001)	(-0.91)	(0.91)	(0.91)	(0.91)	(0.91)	(0.91)	(0.91)	
	(-2.26)**	(-1.48)**	(-0.00)	(0.00)	(0.22)	(1.51)	(0.72)	(-0.21)	(0.80)	(0.87)	
Panel B: Abnormal Levels of and Changes in, Institutional Ownership Sorted by Six Month Prior Performance											
Abnormal log % Inst. (t = 0)	0.0008	0.0119	0.0119	0.0119	0.0091	-0.0091	-0.0071	-0.0091	-0.0091	-0.0091	19.88***
	(0.26)	(3.21)***	(2.86)**	(2.86)**	(2.86)**	(-0.91)	(-2.17)**	(-2.20)**	(-2.20)**	(-2.20)**	
Abnormal Δ % Inst. (t = 11)	-0.0141	-0.0091	-0.0130	-0.0141	-0.0130	-0.0091	-0.0091	-0.0091	-0.0091	-0.0091	17.48***
	(-1.88)**	(-2.48)**	(-3.26)**	(-3.26)**	(-3.26)**	(-1.88)	(-1.88)	(-1.88)	(-1.88)	(-1.88)	
Abnormal Δ % Inst. (t = 0 to 11)	-1.848**	(-1.44)	(-0.20)	(-1.44)	(0.57)	(0.53)	(1.68)	(1.50)	(1.42)	(1.26)	10.33**
	(-2.09)**	(-0.91)	(-0.001)	(-0.91)	(0.91)	(0.91)	(0.91)	(0.91)	(0.91)	(0.91)	

***, **, and * Statistically significant at the 1, 5, and 10 percent levels, respectively.

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Table IV. Momentum Portfolios and Subsequent Changes in Institutional Ownership.

Table VI												
Pre-Herding Year and Herding Year Abnormal Returns by Capitalization Decile												
Each October (1977-1990), NYSE firms are sorted into capitalization deciles. Firms within each capitalization decile are then further sorted into 10 portfolios based on the change in the fraction of shares held by institutional investors over the following year (for a total of 100 capitalization-change in institutional-ownership-sorted portfolios). Reported below are the time-series average of the annual cross-sectional mean abnormal returns for each portfolio. The sample size is 39 annual observations. Abnormal returns are computed by compensating monthly capitalization decile adjusted returns for the period indicated. Panel A presents quarterly abnormal returns for the three months prior to the herding year; Panel B presents annual abnormal returns for the herding year. The F-statistic is based on the null hypothesis that the time-series averages of cross-sectional means do not differ across the ownership change portfolios. Firms must have institutional ownership data at the beginning (t = 0) and end (t = 11) of the herding year and capitalization data at the beginning of the herding year to be included in the sample.												
	% Inst.	Large Decrease	Decile 2	Decile 3	Decile 4	Decile 5	Decile 6	Decile 7	Decile 8	Decile 9	Large Increase	F-stat.
Panel A: Pre-Herding Year Abnormal Three-Month Returns (t = -1 to -3)												
Smallest	0.1600	-0.0090	-0.0007	-0.0731	-0.0337	-0.0002	-0.0199	-0.0246	-0.0218	-0.0161	-0.0373	3.64***
Decile 2	0.2417	-0.0579	-0.0447	-0.0431	-0.0395	-0.0099	-0.0123	-0.0061	-0.0144	0.0042	0.0478	6.24***
Decile 3	0.2940	-0.0351	-0.0119	-0.0011	-0.0113	-0.0111	-0.0085	-0.0041	-0.0061	0.0001	0.0001	0.0001
Decile 4	0.3245	0.0222	-0.0115	-0.0042	0.0306	-0.0071	0.0026	0.0063	0.0003	0.0021	0.0444	3.08***
Decile 5	0.3445	0.0103	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Decile 6	0.3916	0.0053	0.0004	-0.0004	0.0004	0.0003	0.0003	0.0005	0.0005	0.0018	0.0039	0.0245
Decile 7	0.4245	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Decile 8	0.4647	0.0146	0.0132	-0.0005	0.0008	0.0093	0.0050	0.0073	0.0008	0.0125	0.0212	1.22
Decile 9	0.4687	0.0121	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Largest	0.6413	0.0274	0.0255	0.0200	0.0122	0.0041	0.0126	0.0040	0.0047	0.0409	0.0071	1.41
Panel B: Herding Year Annual Abnormal Returns (t = 0 to 11)												
Smallest	-0.1866	-0.1184	-0.1158	-0.0919	-0.0807	-0.0375	-0.0061	0.0558	0.0908	0.2943	0.1678***	
Decile 2	-0.3068	-0.1215	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	
Decile 3	-0.1301	-0.1186	-0.0767	-0.0739	-0.0191	-0.0064	0.0266	0.0408	0.1227	0.0263	0.1678***	
Decile 4	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	
Decile 5	-0.1075	-0.0928	-0.0807	-0.0285	-0.0124	0.0156	0.0420	0.0115	0.0079	0.1568	0.0263	
Decile 6	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	
Decile 7	-0.0610	-0.0308	-0.0020	-0.0546	-0.0770	0.0326	0.0606	0.0271	0.0603	0.0438	0.1372***	
Decile 8	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	
Decile 9	-0.0672	-0.0276	-0.0400	0.0090	-0.0066	-0.0164	0.0171	0.0076	0.0343	0.0805	0.2778***	
Largest	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	-0.0767	

***, ** and * Statistically significant at the 1, 5 and 10 percent levels, respectively.

Panel A: Change in Ownership and Returns for TORQ Sample				
	Large Increase	Small Change	Large Decrease	Increase-Decrease (t-statistic)
Change in ownership	0.0162	0.0023	-0.0106	0.0268 (6.20)***
Raw 3-month return	0.3834	0.1884	0.0610	0.3224 (3.30)***
Abnormal 3-month return	0.1883	0.0113	-0.0996	0.2879 (3.52)***

Panel B: Average Regression Coefficients ($\times 100$)					
β_1 Lag Δ Inst. Ownership	β_2 Return Rank	β_3 Lag Return Rank Measured over the Past k Trading Days			
		1	5	10	20
0.0247	0.0645	-0.0024			
(2.36)**	(7.54)***	(-0.20)			
0.0184	0.0646		0.0182		
(1.66)	(7.63)***		(1.29)		
0.0207	0.0625			0.0062	
(1.76)*	(7.26)***			(0.55)	
0.0197	0.0634				0.0052
(1.87)*	(7.62)***				(0.35)

***, **, and * Statistically significant at the 1, 5, and 10 percent levels, respectively.

Table VII. Returns and Changes in Ownership for TORQ Trader-Type Identified Transactions.

13. Bottom-Line Takeaways

1. Institutional ownership changes are strongly positively related to contemporaneous stock returns.
2. The paper finds no evidence that institutional herding quickly reverses; institutional purchases subsequently outperform institutional sales.
3. Institutions appear to positive-feedback trade, but feedback trading alone does not fully explain the main relation.
4. Institutional ownership changes are closely connected to return momentum.
5. The evidence is consistent with both institutional price pressure and informed institutional trading.
6. The paper is important because it directly connects investor composition, herding, feedback trading, and return momentum using broad ownership data and transaction-level evidence.